

Label Studio Setup for Color Grid Annotation

You have been given two files:

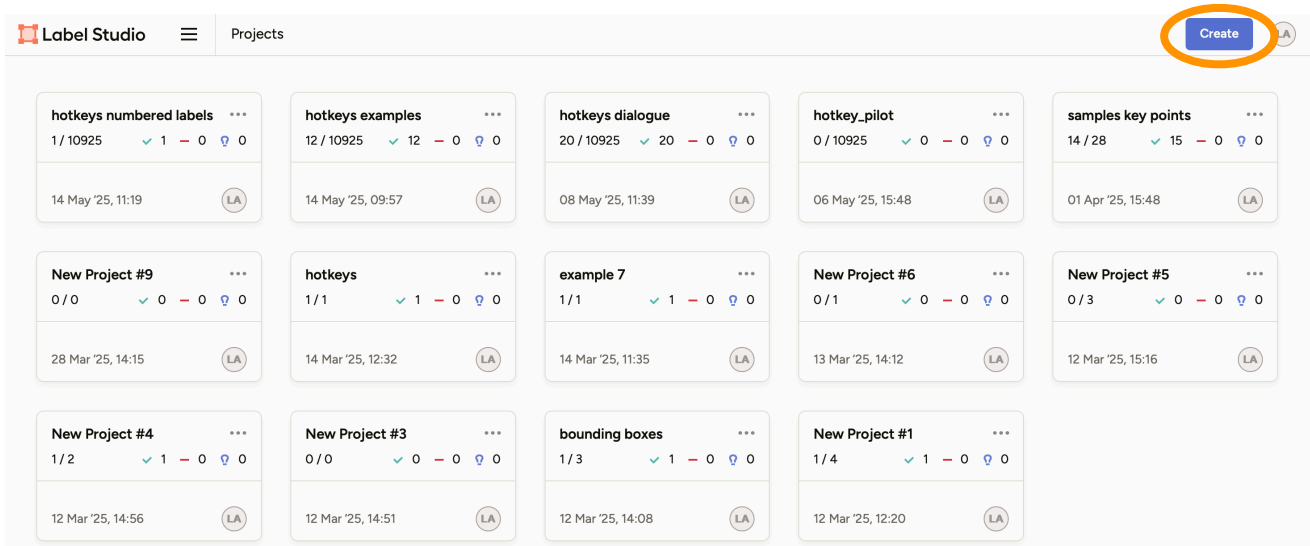
- labeling_interface.xml (Code for label studio to provide the graphic interface)
- pilotnumber_letter.json (e.g. pilot1_a (numbers and letters are varying) containing the data)

1. Install label studio

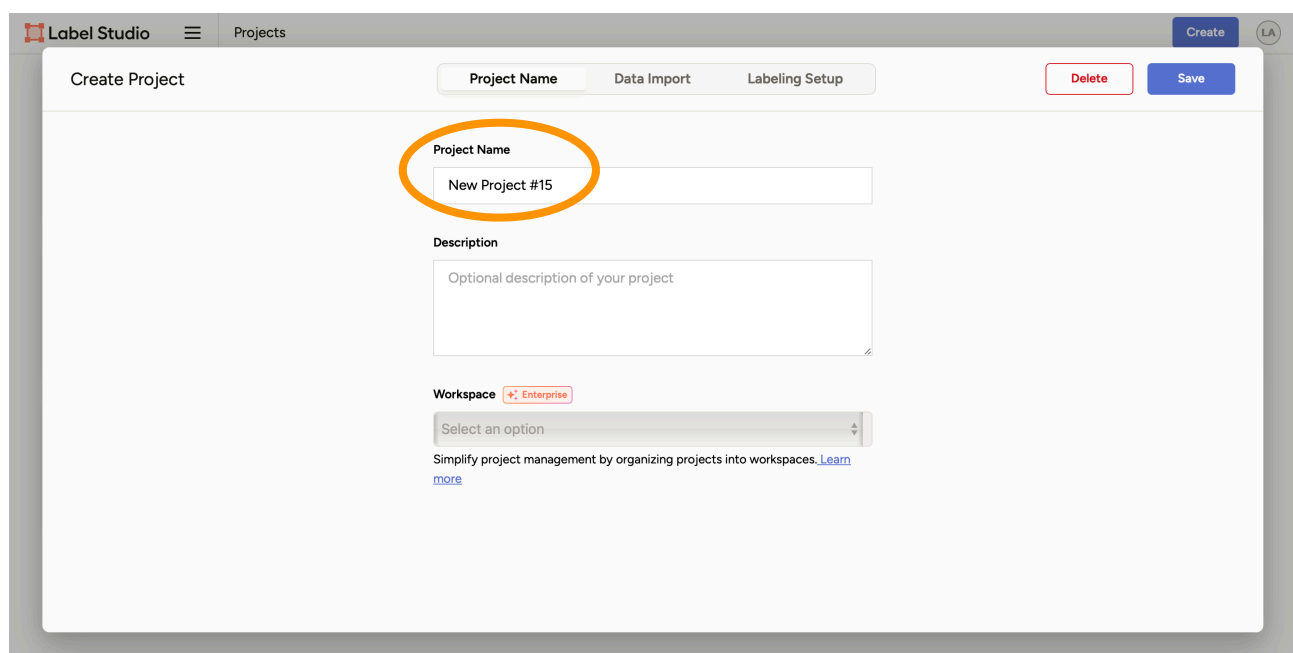
Follow the instructions on <https://labelstud.io/> for quick start. After installing and starting label studio you will have to sign up with your email and create a password.

2. Create a new label studio annotation project

Once label studio is installed and runs on <http://localhost:8080> create a new project.

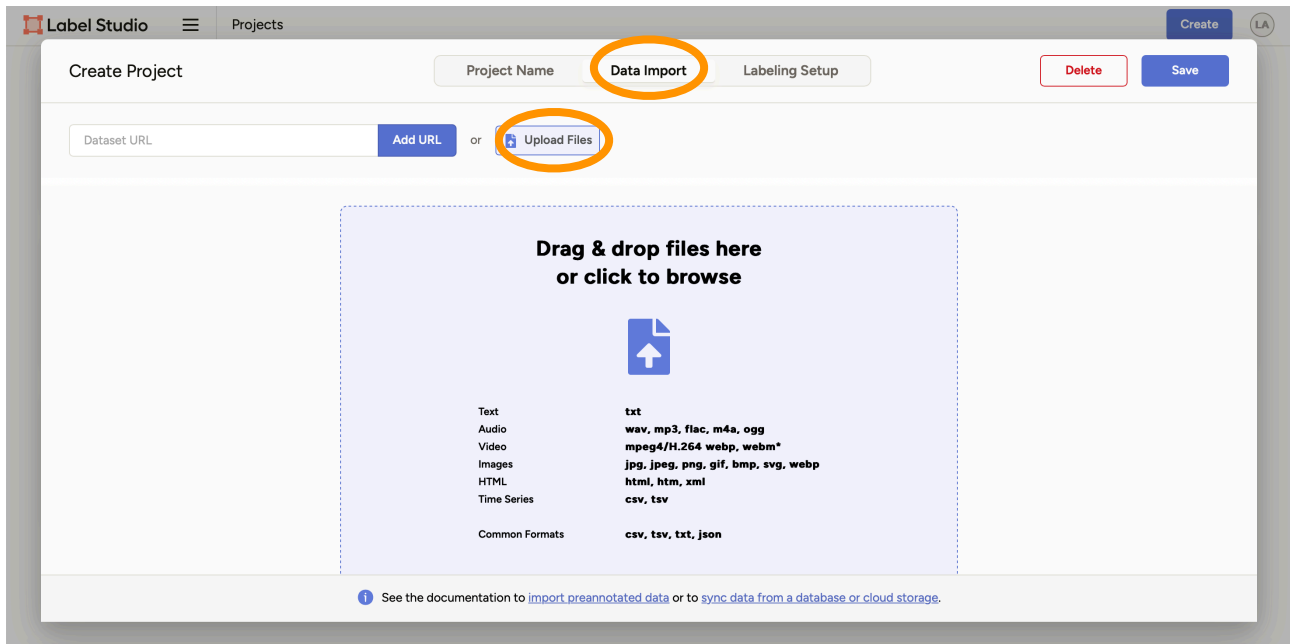


Alter the project name to your specific json file name (if your provided file is pilot1_a.json the label studio project name should be **pilot1_a**)



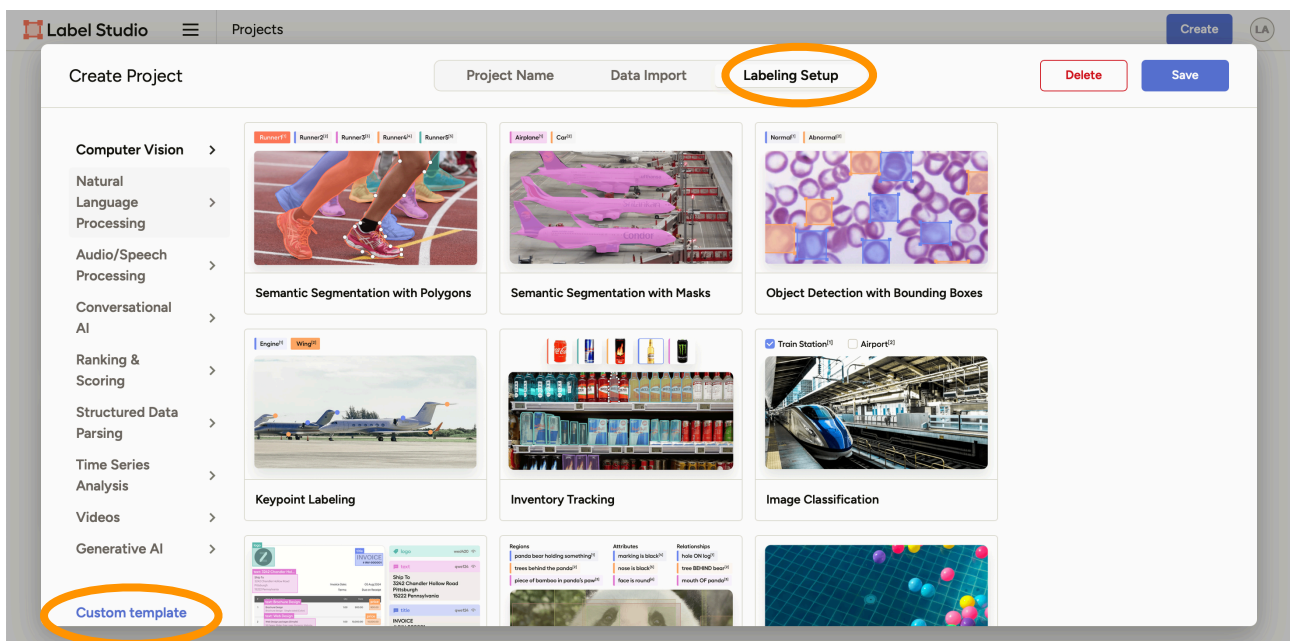
3. Import data

Click on the „Data Import“ tab after altering the project name. Use the „Upload Files“ button or drag & drop your specific json file (e.g. pilot1_a.json) here.

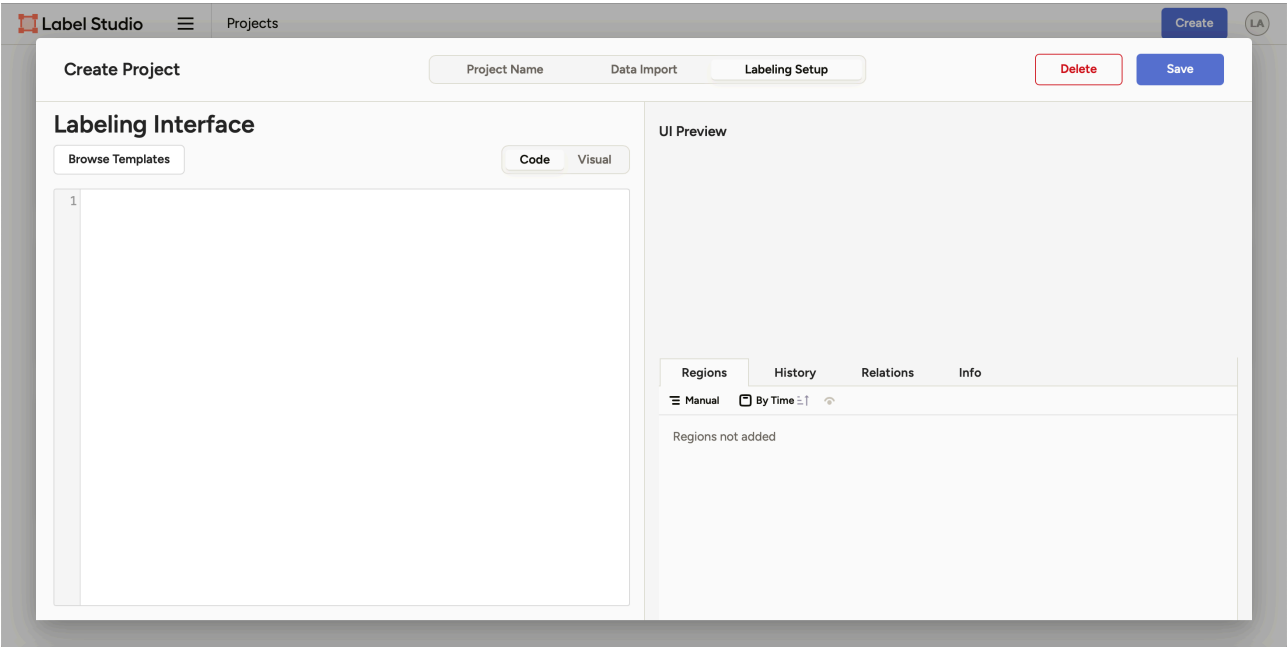
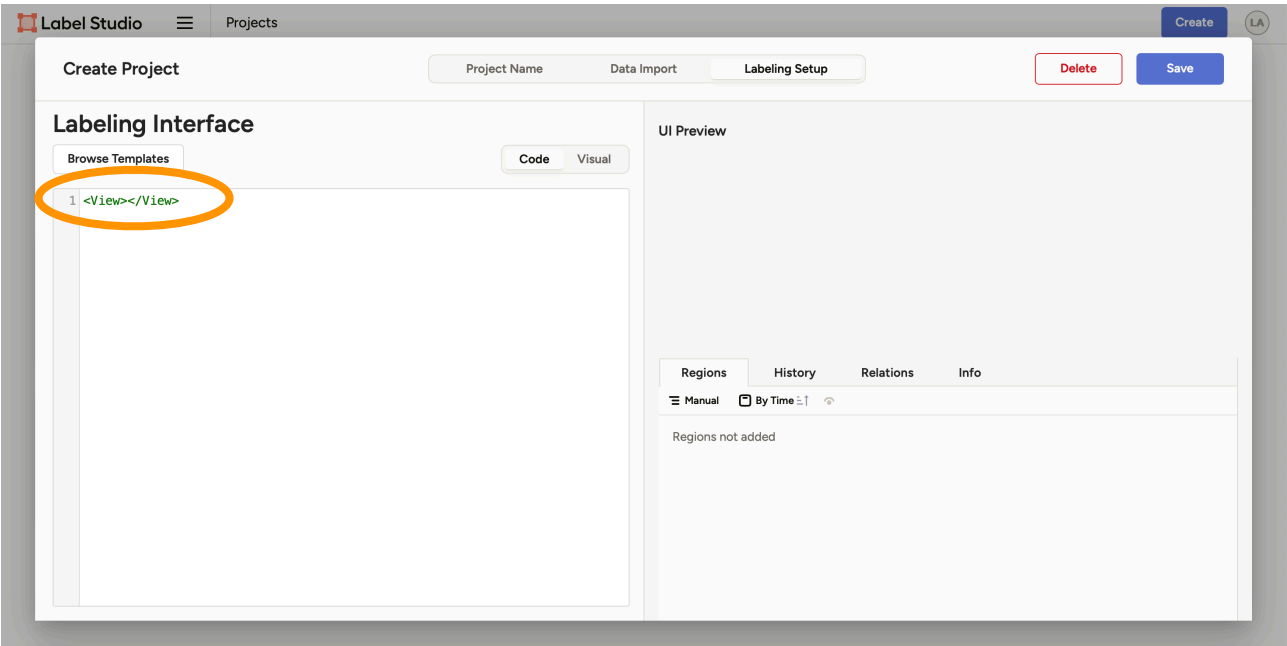


4. Setup Labelling interface

Click on the „Labeling Setup“ tab on top and then on „Custom template“ at the bottom left.



Delete `<View></View>` to completely empty the labeling interface.



The screenshot shows the Label Studio web application. At the top, there's a navigation bar with 'Label Studio', a hamburger menu, 'Projects', and a 'Create' button. Below this is a 'Create Project' dialog with three tabs: 'Project Name', 'Data Import', and 'Labeling Setup'. The 'Labeling Setup' tab is active, showing a 'Delete' button and a 'Save' button, which is circled in red. Below the dialog is the 'Labeling Interface'. It has a 'Browse Templates' button and two tabs: 'Code' and 'Visual'. The 'Code' tab is selected, showing XML-like code for defining a labeling interface. The code includes a <View> element with <Image>, <Text>, and <Labels> sub-elements. The <Labels> element defines three label types: 'positive' (green background), 'negative' (red background), and 't1' (pink background). The 'Visual' tab shows a preview of the labeling interface. It features an aerial image of an airport tarmac with several aircraft. Bounding boxes are drawn around the aircraft, and labels are assigned to them. The labels are 'positive', 'negative', and 't1'. The 't1' labels are further categorized into 'Target', 'Distractor1', and 'Distractor2'. The 'Target' labels are 't1', 't2', and 't3'. The 'Distractor1' labels are 'd1.1', 'd1.2', 'd1.3', 'd1.4', 'd1.5', 'd1.6', 'd1.7', 'd1.8', and 'd1.9'. The 'Distractor2' labels are 'd2.1', 'd2.2', 'd2.3', 'd2.4', 'd2.5', 'd2.6', 'd2.7', 'd2.8', and 'd2.9'. The 't1' labels are highlighted in pink, and the 'Distractor1' and 'Distractor2' labels are highlighted in light blue.

Then you will see your 20 annotation tasks. You can start by clicking on one of the tasks.

Label Studio

Projects / pilot1_a

Settings

LA

Default

Actions

Columns

Filters

not set

Label All Tasks

↺

Import

Export

List

Grid

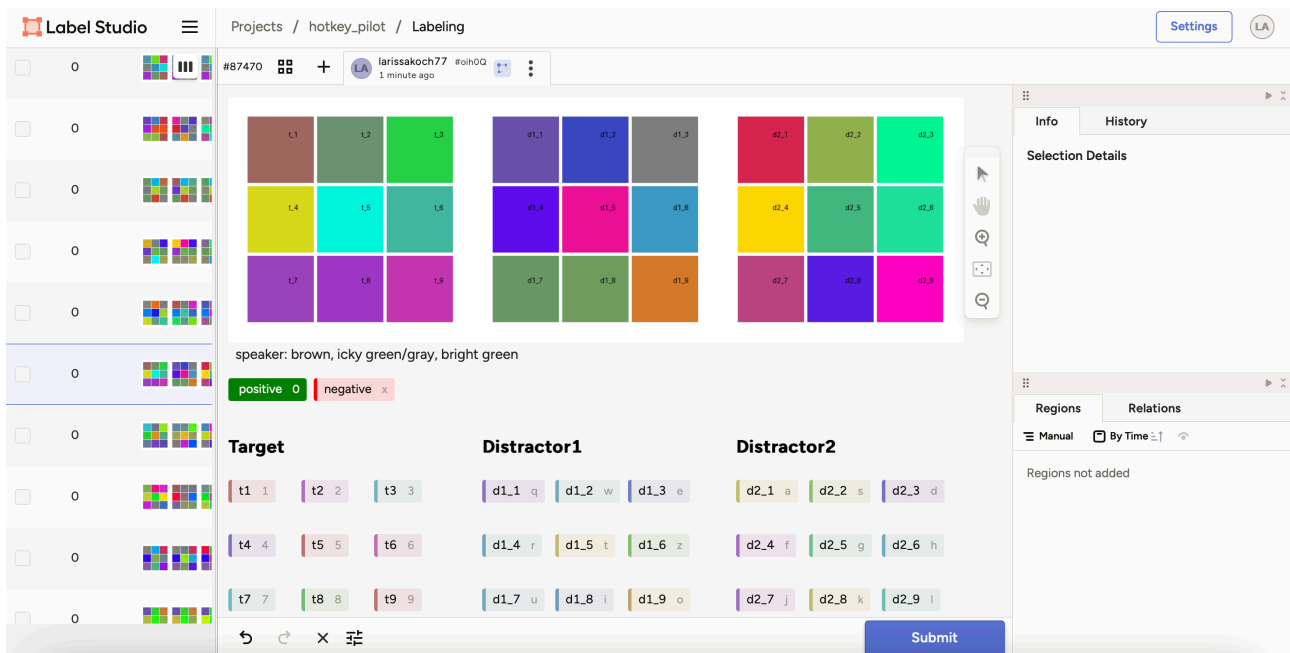
ID	Completed				Annotated by	url	img	text	str
87444	0	0	0				listener: maybe im just colorblind speaker: sorry looked tan listener: its my		
87445	0	0	0				speaker: Purple upper right		
87446	0	0	0				speaker: middle is bright pink color		
87447	0	0	0				speaker: the left middle square is the brightest pink		
87448	0	0	0				speaker: upper right is aqua		
87449	0	0	0				listener: One of the two identical ones? speaker: yes listener: Um, okay :?O		
87450	0	0	0				listener: back to easy? haha speaker: haha speaker: Ok bottom middle is gold		
87451	0	0	0				speaker: center bright		

Tasks: 20 / 20

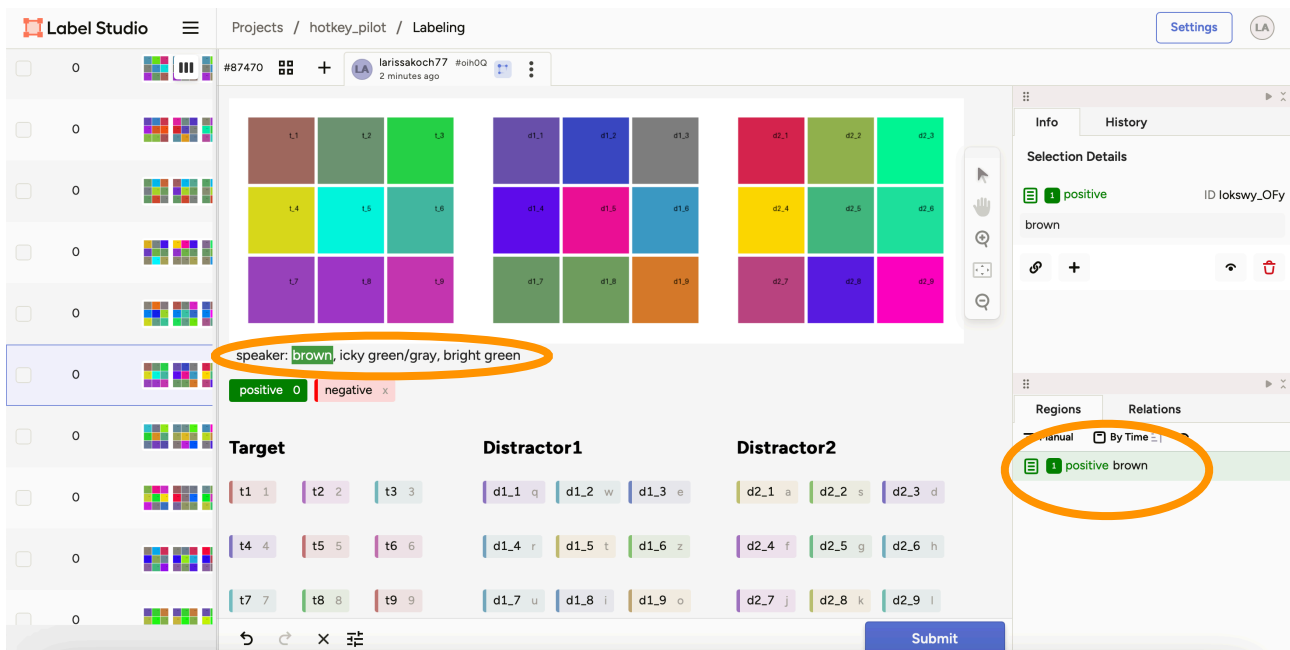
Submitted annotations: 0

Predictions: 0

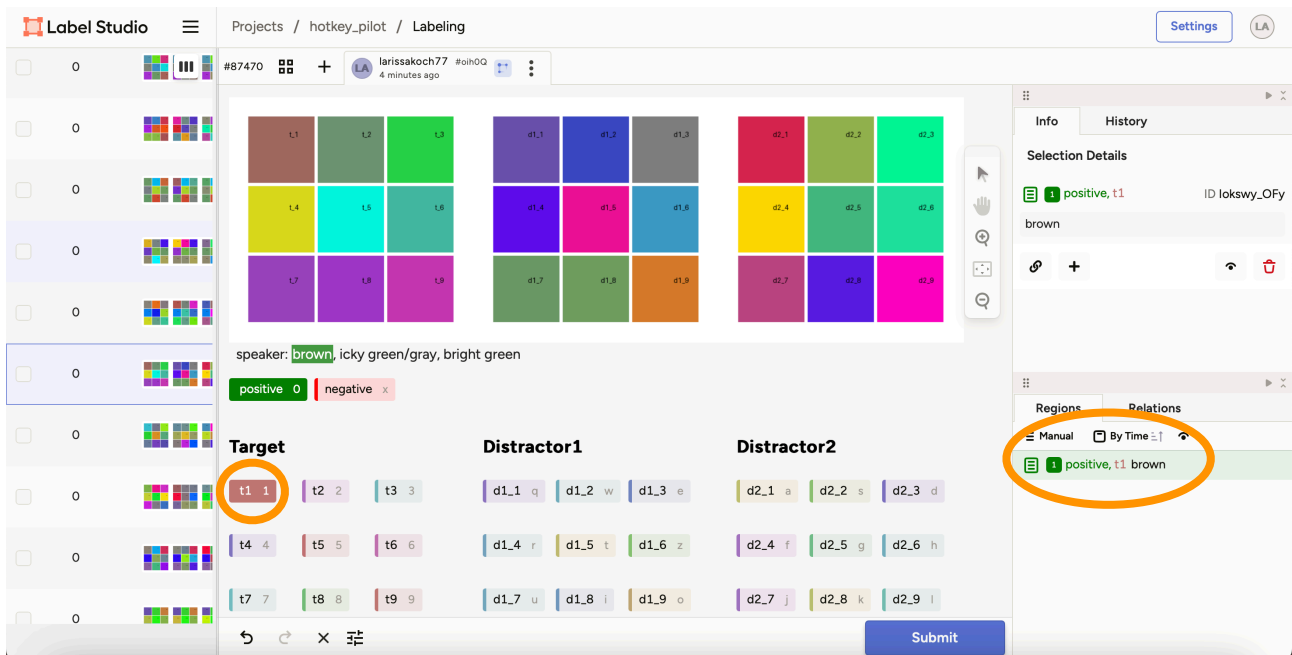
For each annotation item you will see the image on the top always in the order „Target, Distractor1, Distractor2“. Below the image are the corresponding utterances of this round. They can also contain listener utterances and might even be short dialogues between the participants. Below the utterances are the two labels „positive“ and „negative“. The positive label will always be selected by default since it is more common. It can be deselected by clicking on it to then select the negative label in cases where needed. See annotation guidelines for the use of the „negative“ label.



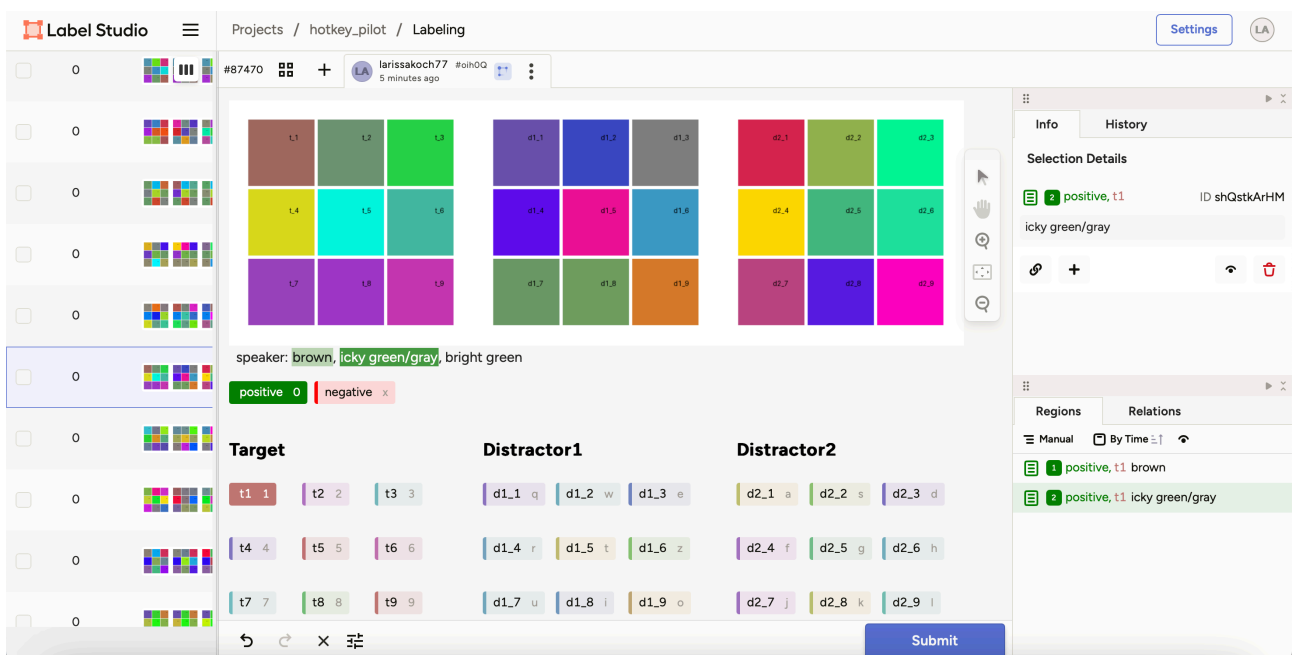
When the correct label („positive“ or „negative“) is selected either by default or by manually selecting, you can mark the relevant text span just on the shown text itself which will then automatically be labeled with either positive or negative.



Now combine the labeled text span with the relevant part of the image by choosing the corresponding hotkey. Either by clicking on the visible hotkeys (marked in the image below) or by pressing the actual key on your keyboard (in this example 1). Now you can see the text span, positive/negative label and the hotkey label (t1 in this case) on the right bottom.

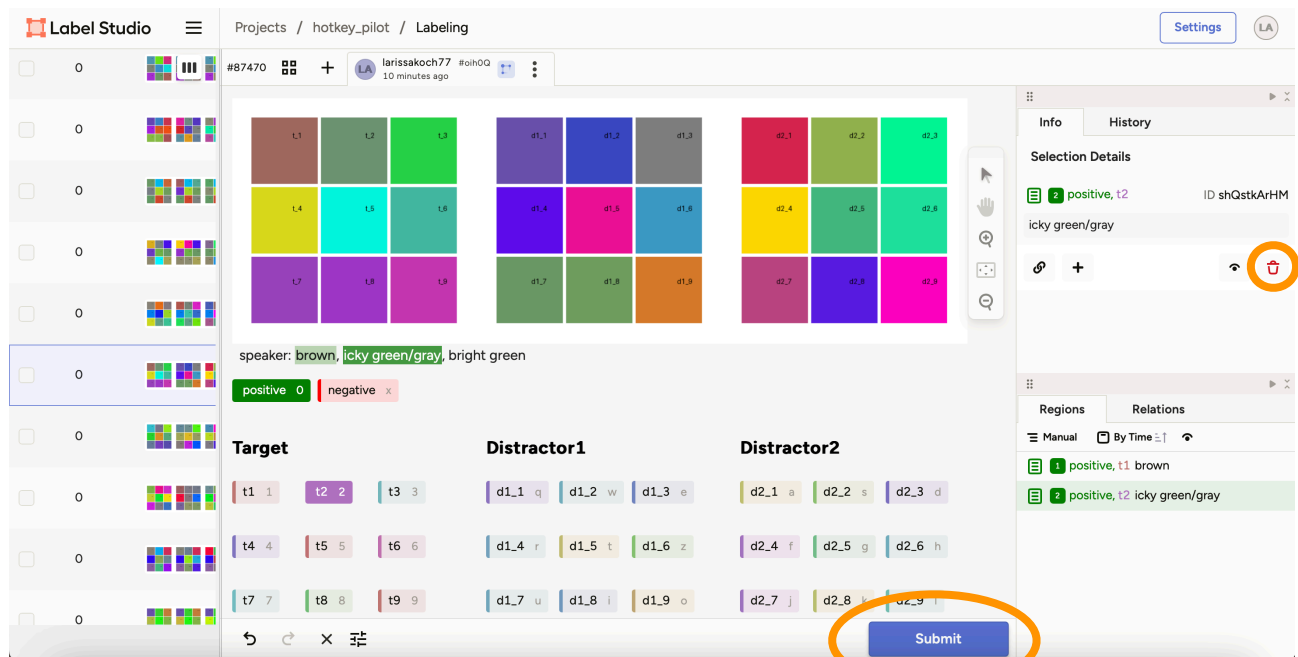


You can continue by marking another new span when needed. A new label with the new text span will be created. CAUTION: the hotkey from the previous span will stay activated. You have to deselect it by clicking on it or pressing the related key on your keyboard. Then you select the new hotkey.



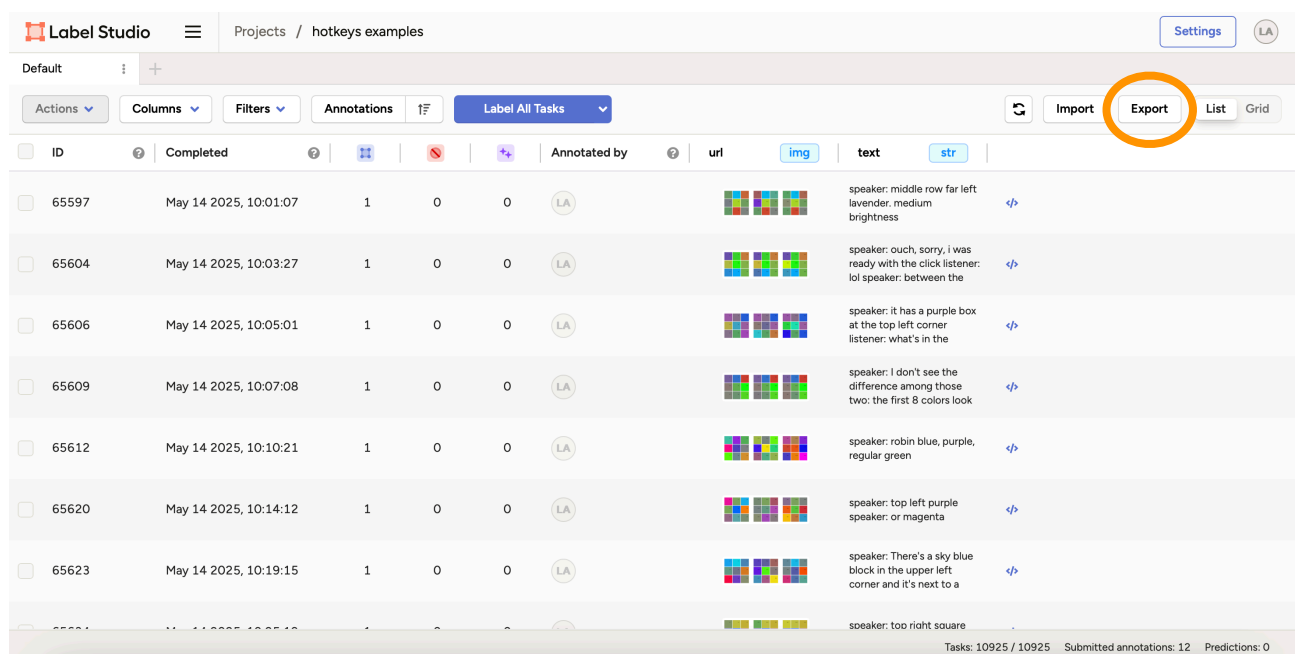
You can always check in the right bottom corner if the assignment of positive/negative label and hotkeys (t or d1/d2 labels) is as you want it to be for each text span. If something gets mixed up you have the opportunity to select the wrong annotation and delete it to start again by marking the text span and selecting the desired hotkeys.

When you are ready with the annotation for the shown round click on „Submit“

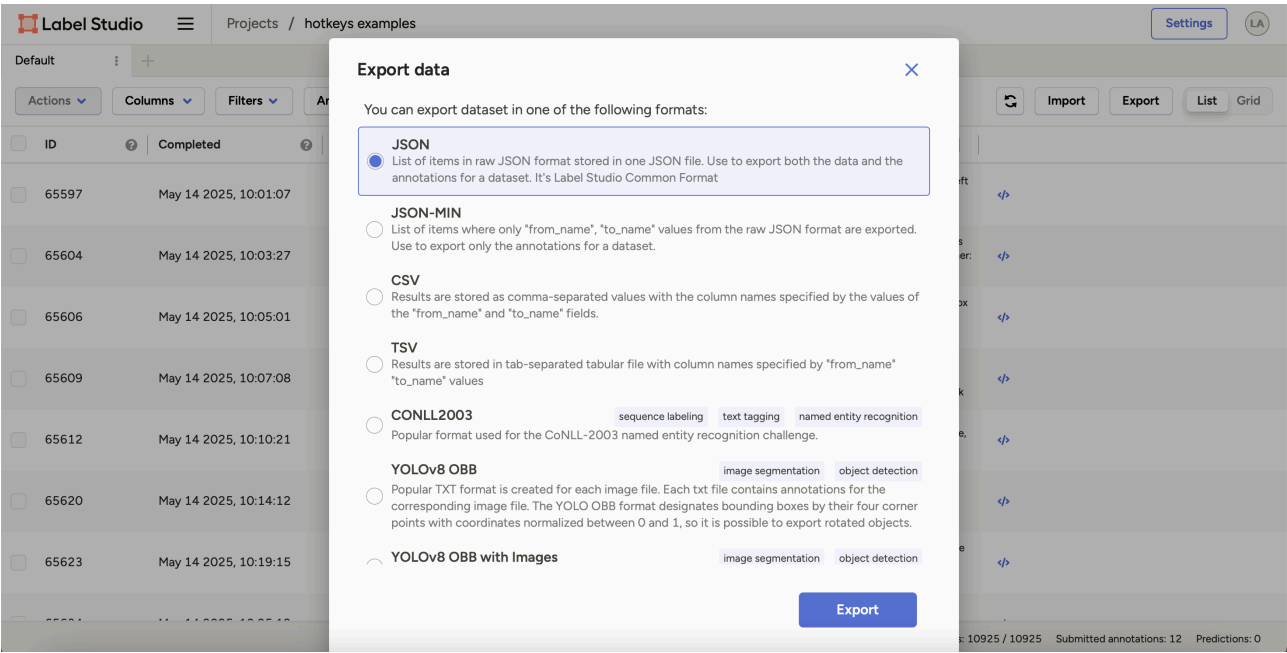


6. Export the annotations

Once you are finished with the annotations click on „Export“.



Export your annotations as a json file.



7. Annotations guidelines